

# Description and choice data for the domain “Cars”

## Description of the choice domain 23, Cars

The prompt question and the universe of five response options in the choice domain **Cars** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

This domain is based on an experiment from Wedell and Pettibone (1996). There are two experiments involving cars in that article, the experiment in question is numbered 18 in the appendix to that paper. Objects below have the same attributes as in that experiment and a similar range of levels. We adapted the objects to allow for compromise effects. The second panel of Figure ?? shows the choice objects in attribute space. Table ?? shows the relations of dominance, similarity and betweenness among objects associated with context effects.

Which one of the following cars would you choose to drive, all other features begin equal? Ride quality is a on a scale of 0 to 100.

| Ride quality | Miles per gallon |
|--------------|------------------|
| 60           | 30               |
| 80           | 24               |
| 70           | 27               |
| 55           | 28               |
| 75           | 22               |

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

**Cars**

Which one of the following similarly priced cars would you choose to drive, based on the percentage of owners who indicated that they “would buy again,” and the fuel economy rating?

|                       | Percentage responding “would buy again” | Fuel economy rating (a lower rating means more fuel efficient) in litres per 100km |
|-----------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| <input type="radio"/> | 75.0                                    | 9.5                                                                                |
| <input type="radio"/> | 65.0                                    | 7.5                                                                                |
| <input type="radio"/> | 72.5                                    | 10.0                                                                               |

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Figure 1: Screenshot for domain Cars

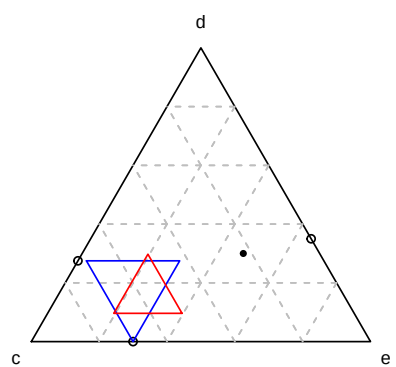
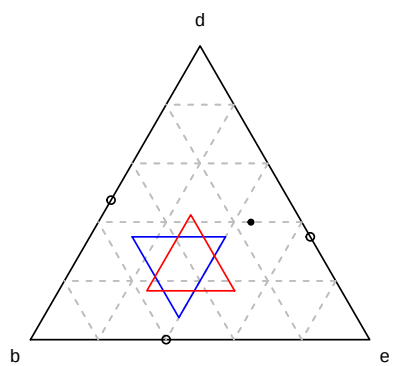
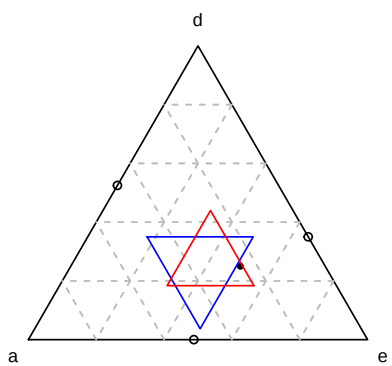
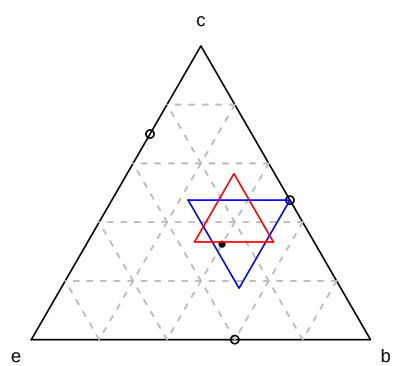
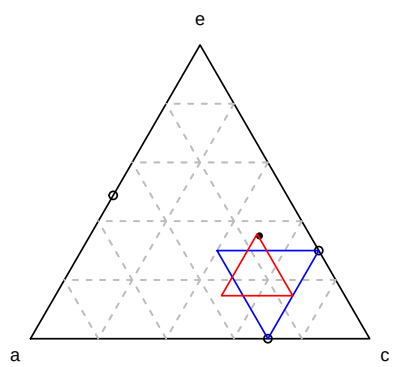
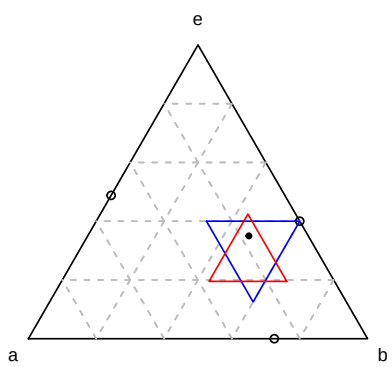
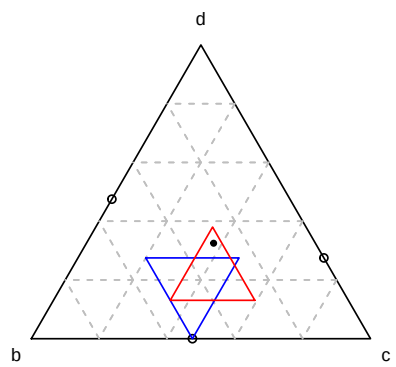
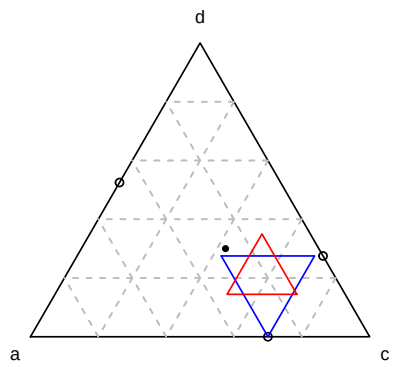
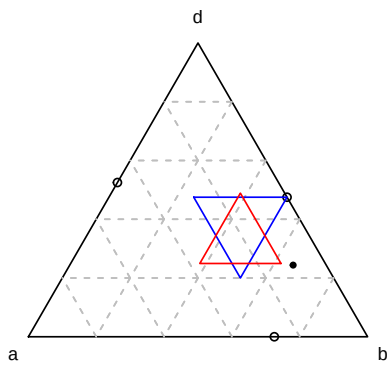
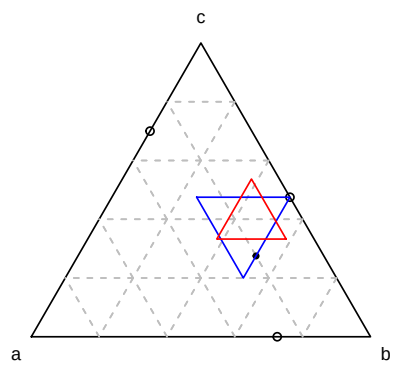
| Menu $A$            | Choice counts |          |          |          |          | Choice proportions |                |                |                |                |
|---------------------|---------------|----------|----------|----------|----------|--------------------|----------------|----------------|----------------|----------------|
|                     | $N_A(a)$      | $N_A(b)$ | $N_A(c)$ | $N_A(d)$ | $N_A(e)$ | $\hat{P}_A(a)$     | $\hat{P}_A(b)$ | $\hat{P}_A(c)$ | $\hat{P}_A(d)$ | $\hat{P}_A(e)$ |
| $\{a, b\}$          | 11            | 29       | -        | -        | -        | 0.275              | 0.725          | -              | -              | -              |
| $\{a, c\}$          | 12            | -        | 28       | -        | -        | 0.300              | -              | 0.700          | -              | -              |
| $\{b, c\}$          | -             | 21       | 19       | -        | -        | -                  | 0.525          | 0.475          | -              | -              |
| $\{a, b, c\}$       | 8             | 21       | 11       | -        | -        | 0.200              | 0.525          | 0.275          | -              | -              |
| $\{a, d\}$          | 19            | -        | -        | 21       | -        | 0.475              | -              | -              | 0.525          | -              |
| $\{b, d\}$          | -             | 21       | -        | 19       | -        | -                  | 0.525          | -              | 0.475          | -              |
| $\{a, b, d\}$       | 4             | 27       | -        | 10       | -        | 0.098              | 0.659          | -              | 0.244          | -              |
| $\{c, d\}$          | -             | -        | 29       | 11       | -        | -                  | -              | 0.725          | 0.275          | -              |
| $\{a, c, d\}$       | 11            | -        | 17       | 12       | -        | 0.275              | -              | 0.425          | 0.300          | -              |
| $\{b, c, d\}$       | -             | 12       | 15       | 13       | -        | -                  | 0.300          | 0.375          | 0.325          | -              |
| $\{a, b, c, d\}$    | 2             | 20       | 10       | 8        | -        | 0.050              | 0.500          | 0.250          | 0.200          | -              |
| $\{a, e\}$          | 21            | -        | -        | -        | 20       | 0.512              | -              | -              | -              | 0.488          |
| $\{b, e\}$          | -             | 24       | -        | -        | 16       | -                  | 0.600          | -              | -              | 0.400          |
| $\{a, b, e\}$       | 7             | 19       | -        | -        | 14       | 0.175              | 0.475          | -              | -              | 0.350          |
| $\{c, e\}$          | -             | -        | 28       | -        | 12       | -                  | -              | 0.700          | -              | 0.300          |
| $\{a, c, e\}$       | 6             | -        | 20       | -        | 14       | 0.150              | -              | 0.500          | -              | 0.350          |
| $\{b, c, e\}$       | -             | 16       | 13       | -        | 11       | -                  | 0.400          | 0.325          | -              | 0.275          |
| $\{a, b, c, e\}$    | 6             | 14       | 13       | -        | 7        | 0.150              | 0.350          | 0.325          | -              | 0.175          |
| $\{d, e\}$          | -             | -        | -        | 14       | 26       | -                  | -              | -              | 0.350          | 0.650          |
| $\{a, d, e\}$       | 10            | -        | -        | 10       | 20       | 0.250              | -              | -              | 0.250          | 0.500          |
| $\{b, d, e\}$       | -             | 6        | -        | 16       | 18       | -                  | 0.150          | -              | 0.400          | 0.450          |
| $\{a, b, d, e\}$    | 3             | 17       | -        | 4        | 16       | 0.075              | 0.425          | -              | 0.100          | 0.400          |
| $\{c, d, e\}$       | -             | -        | 9        | 12       | 19       | -                  | -              | 0.225          | 0.300          | 0.475          |
| $\{a, c, d, e\}$    | 8             | -        | 16       | 6        | 10       | 0.200              | -              | 0.400          | 0.150          | 0.250          |
| $\{b, c, d, e\}$    | -             | 9        | 10       | 8        | 13       | -                  | 0.225          | 0.250          | 0.200          | 0.325          |
| $\{a, b, c, d, e\}$ | 4             | 11       | 7        | 14       | 4        | 0.100              | 0.275          | 0.175          | 0.350          | 0.100          |

Table 1: Observed choice counts and proportions.

## Choice data for domain 23, Cars

Table 1 shows choice counts and choice proportions for this choice domain. For each menu  $A$  and each object  $x \in \{a, b, c, d, e\}$ ,  $N_A(x)$  is the number of participants who chose object  $x$  from menu  $A$  and  $\hat{P}_A(x)$  is the corresponding proportion of participants who chose  $x$  from  $A$ . When  $x \notin A$ , a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of  $\{a, b, c, d, e\}$ . The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.



## References

Douglas H. Wedell and Jonathan C. Pettibone. Using judgments to understand decoy effects in choice. *Organization Behavior and Human Decision Processes*, 67(3):326—344, 1996.